Find a Friend: **Leaf Structure and Function Activity**

Teacher Instructions

This activity is good for consolidating or checking learning in a plenary or for revision.
Cards can be printed out A4 or A5 and laminated for repeated use.

How to use:

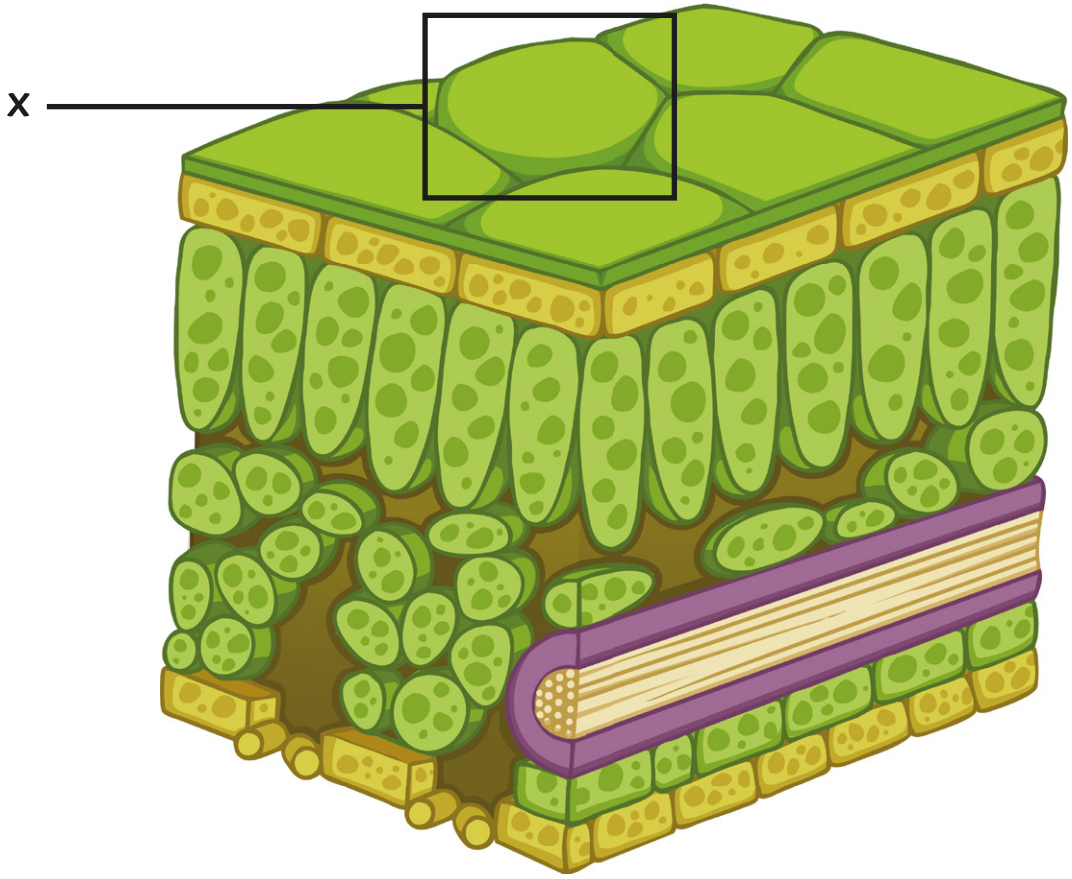
There are 3 types of card:

- orange – diagram (with a part labelled as X);
- blue – name of a part;
- green – description or function of a part.

Give each student in the class one of the cards. This can be random or differentiated by giving easier cards to weaker students and more tricky cards to higher-ability students. Unused cards can be put onto a table for students to go to if they can't find their 'friends'. Where there are more students than cards, give one card to a pair of students.

Students should then move around the room and find their two 'friends' with the matching diagram, name and function. This encourages discussion and students that are unsure can get help from their peers.

Once the students have found their friends, the cards can be stuck onto a board, wall or window to form a display that students can refer to if necessary.



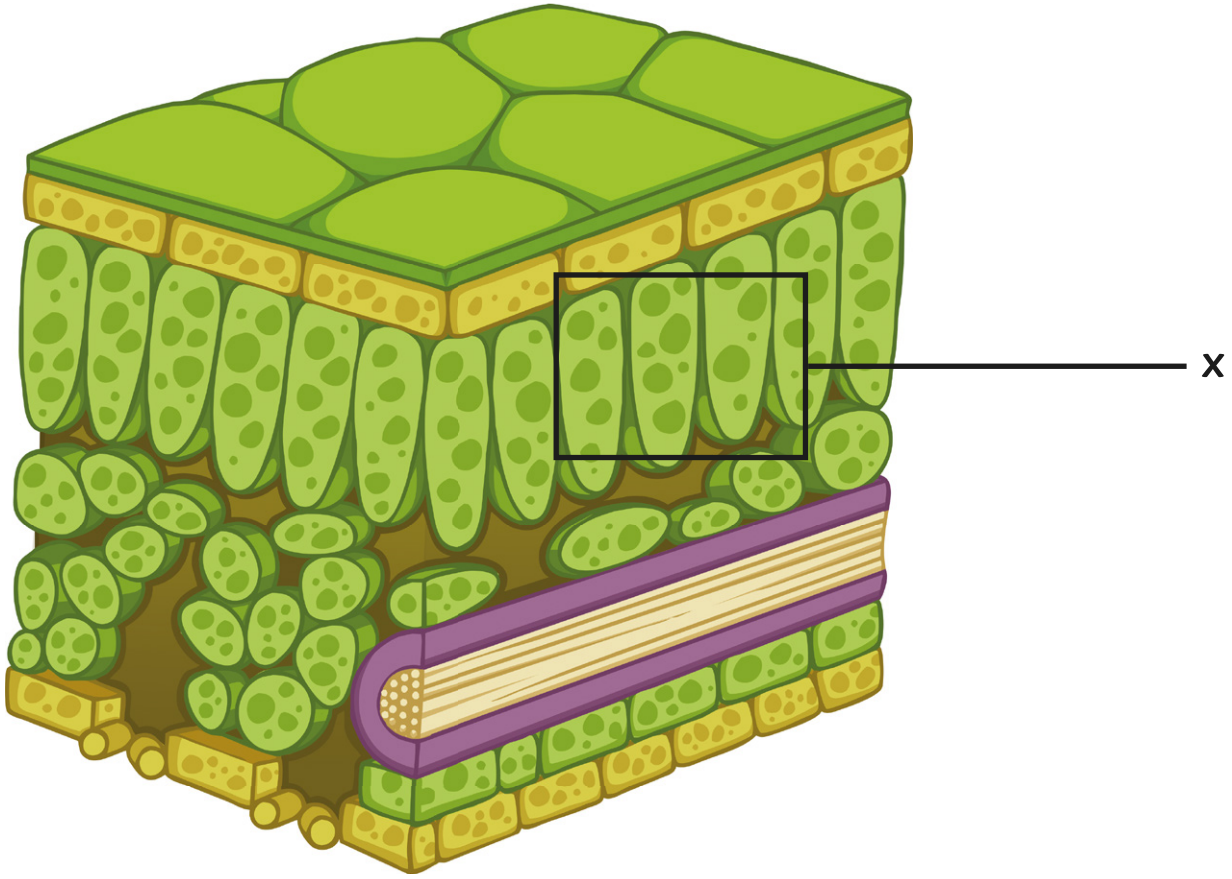
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waxy cuticle

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A water-repellent layer that protects the leaf and reduces water loss from the leaf.

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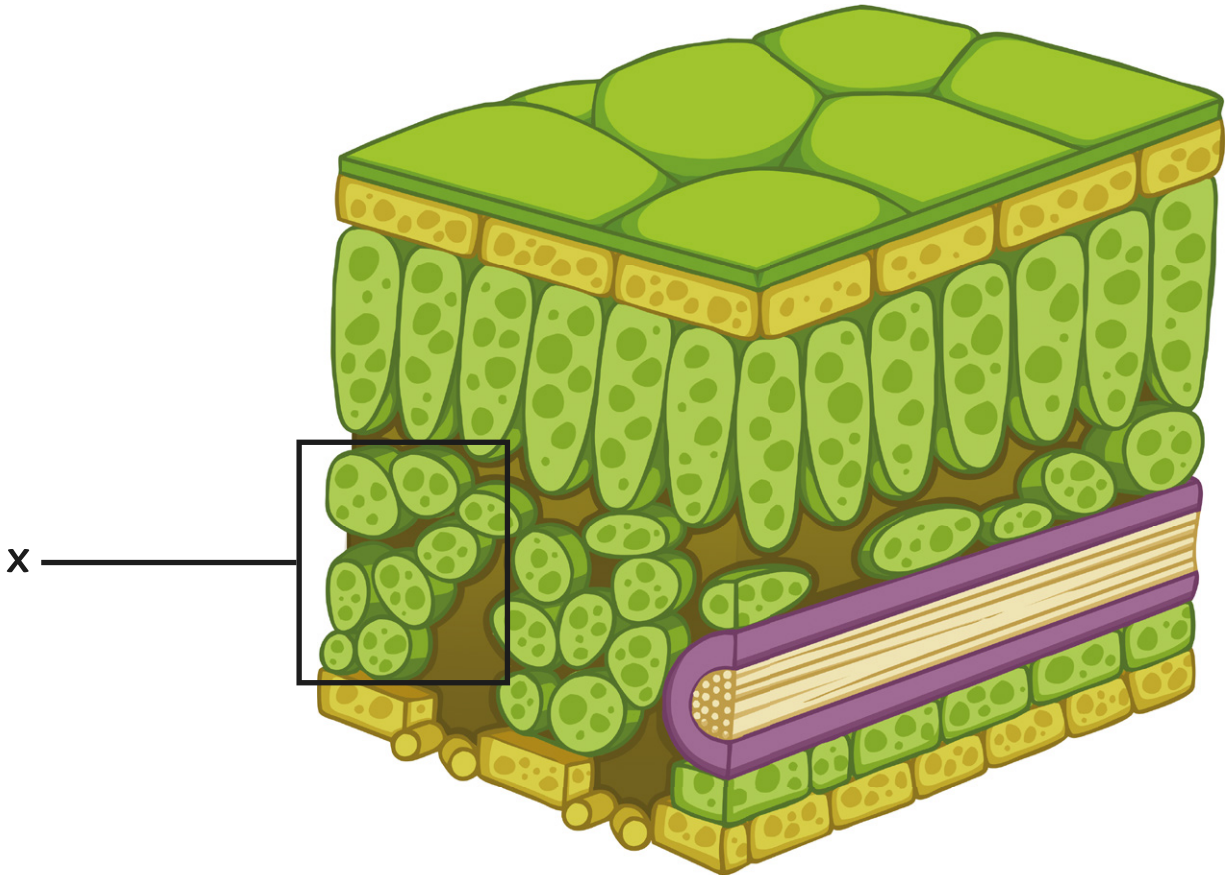
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palisade layer

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This is made of palisade cells, which have a large surface area and contain lots of chloroplasts to absorb sunlight for photosynthesis.

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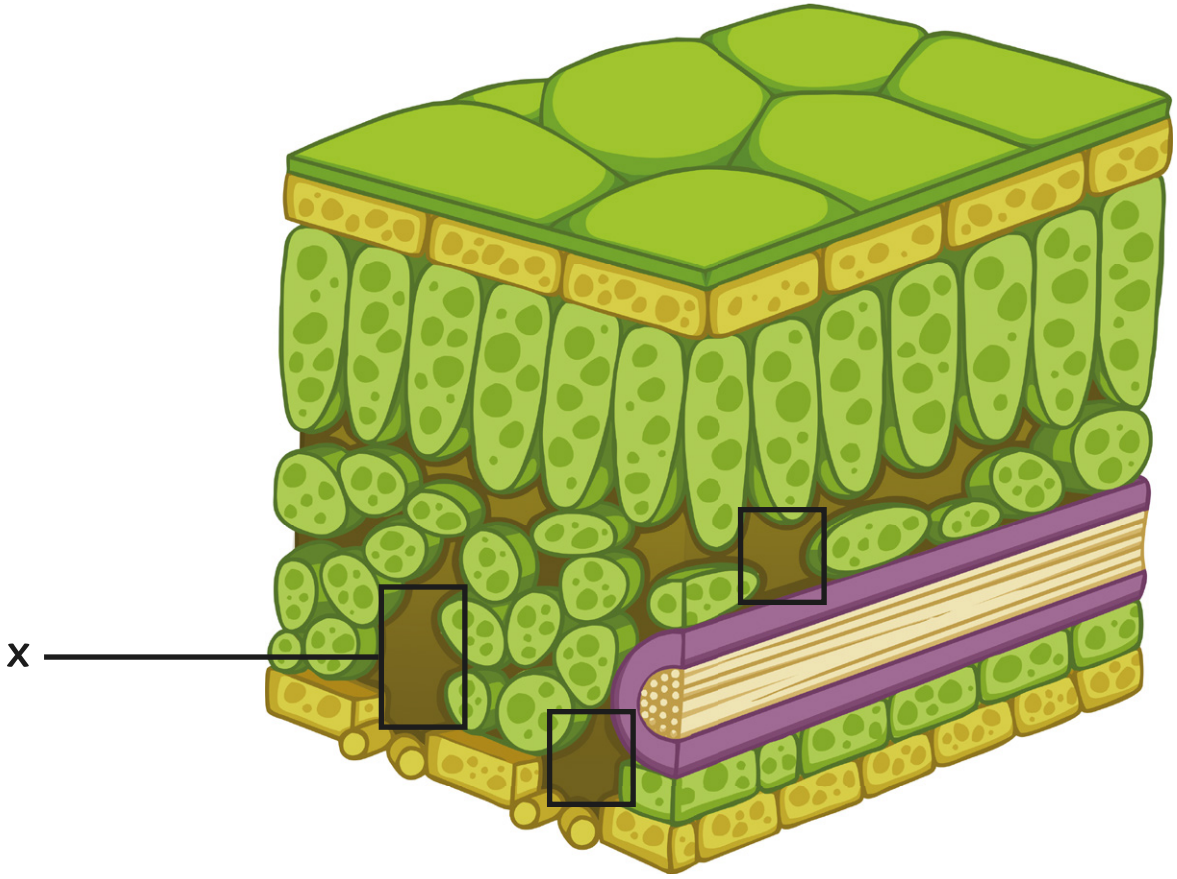
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spongy mesophyll

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A layer of loose-fitting cells with many air spaces to allow gases to diffuse easily.

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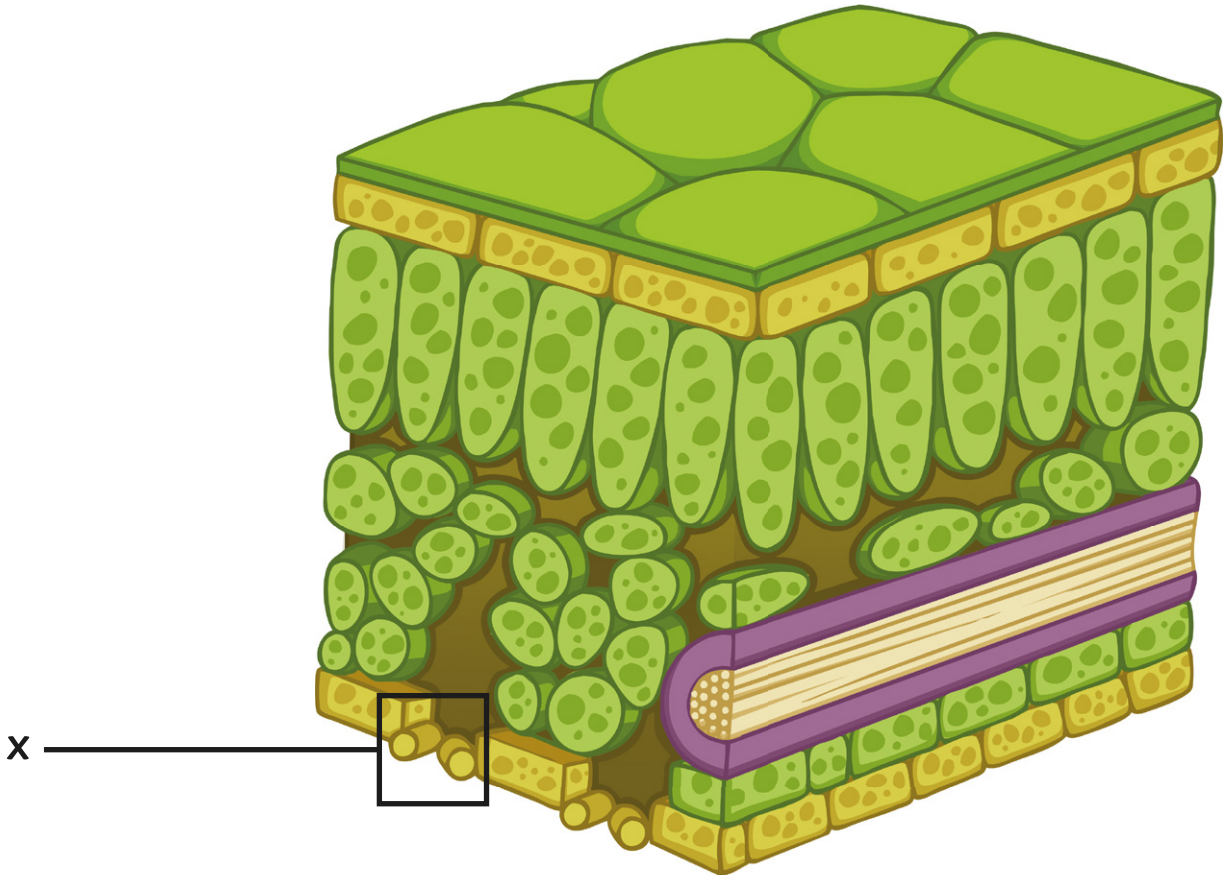
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air space

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A space that allows gases to be exchanged easily.

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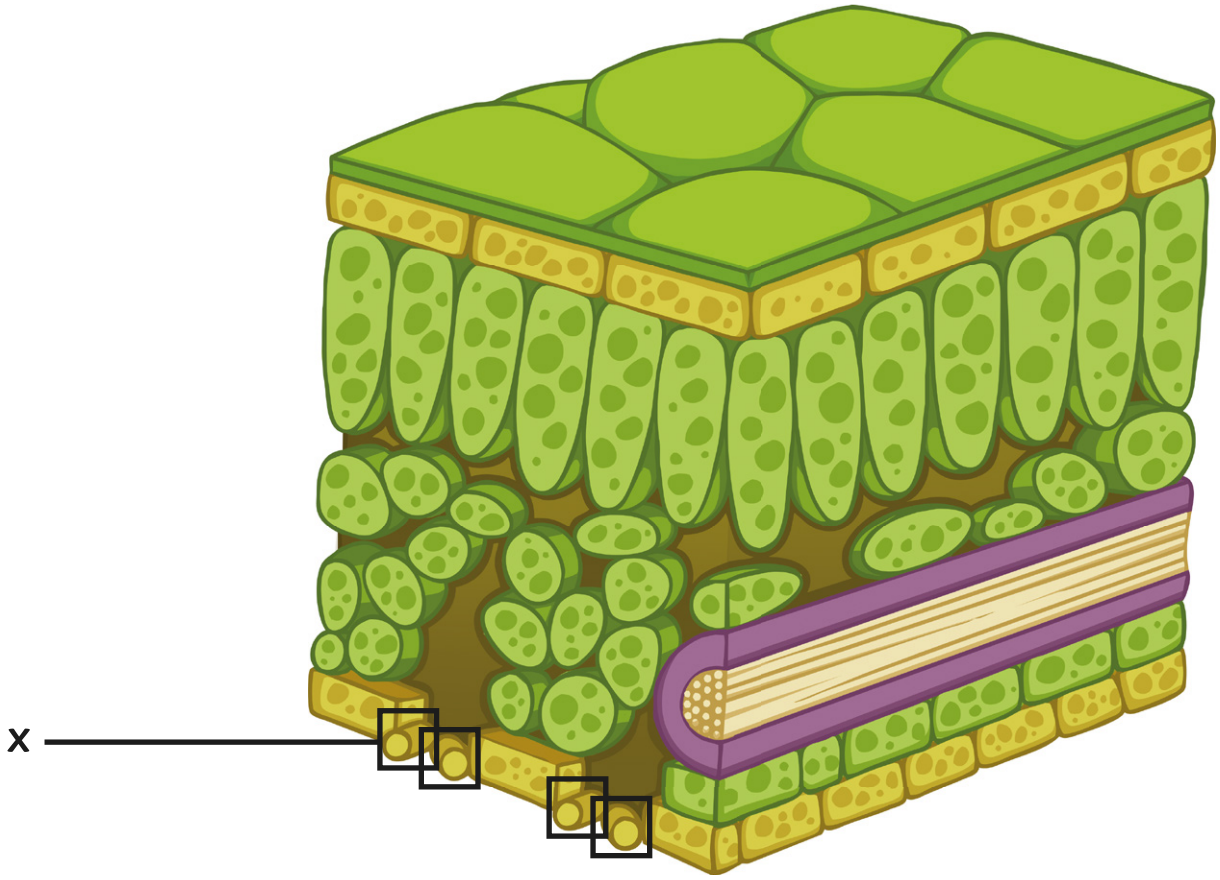
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stoma

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A pore that allows gases to pass into and out of the leaf.

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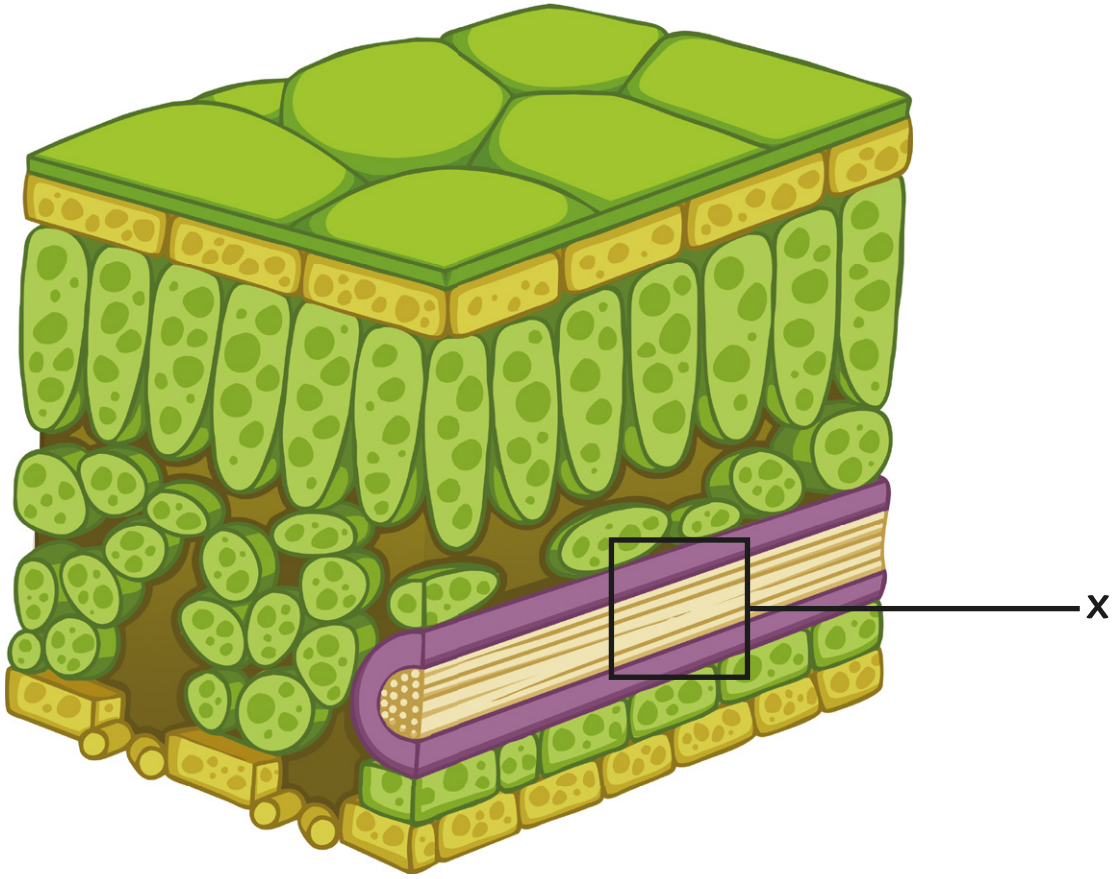
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guard cells

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These open and close pores in the leaf to control water loss.

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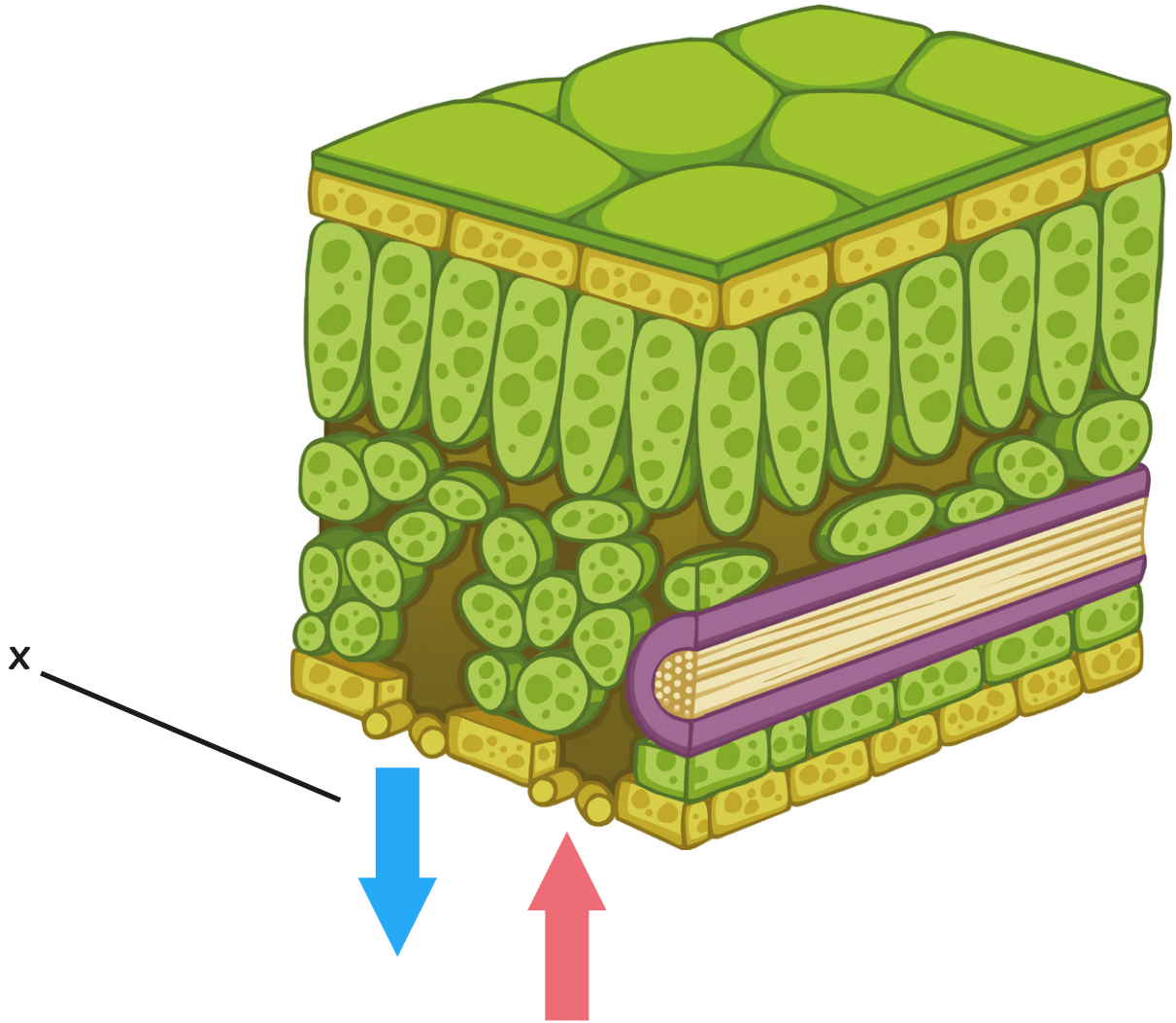
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leaf vein

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This contains xylem and phloem tubes that transport water, minerals and sugars to and from the leaf.

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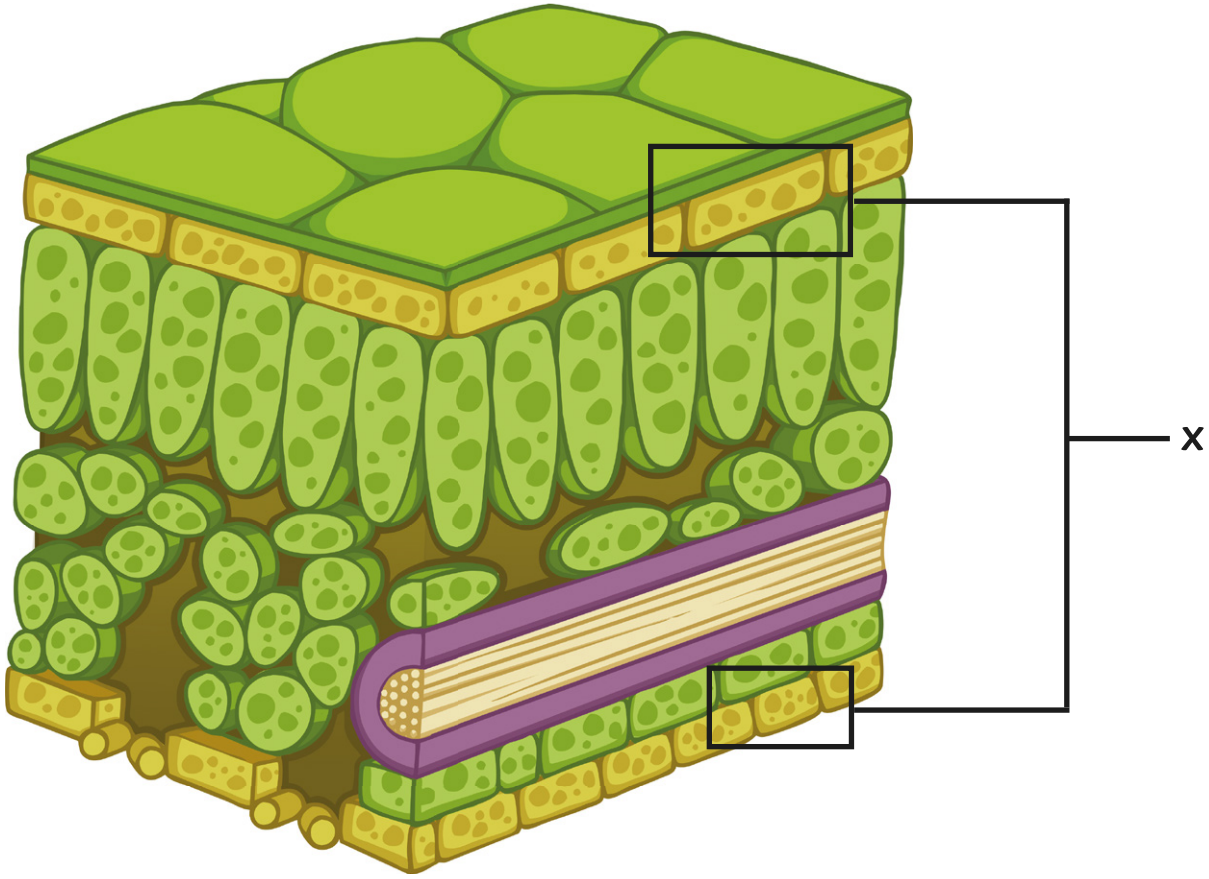
gaseous exchange

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The diffusion of:

- carbon dioxide into the leaf (for photosynthesis) and;
- oxygen and water vapour out of the leaf.

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epidermis

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A transparent layer of tightly-packed cells that protects the leaf.

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